SAINT MICHAEL COLLEGE OF CARAGA

FOOD SAFETY AND SANITATION PRACTICES AMONG LOCAL BAKERIES IN SELECTED MUNICIPALITIES OF AGUSAN DEL NORTE: BASIS FOR POLICY ENHANCEMENT

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By

ANGEL MAE YONSON
PRINCESS MAY JUSSILY B. BATANG
LESTER VONN U. BLANCO
ROSEMARIE F. BULLO
ALTHEA MONICA N. MONTILLA
BRYAN B. MUCA

ADVISER: MA. LILIAN PACE, MM
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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

Food safety and sanitation remain critical public health concerns worldwide due to their direct impact on human health and economic stability. According to the World Health Organization (WHO, 2024), approximately 600 million people, nearly one in ten globally, experience foodborne illnesses each year, resulting in about 420,000 deaths, including 125,000 children under the age of five. Beyond health consequences, unsafe food practices contribute to substantial economic losses, particularly in low- and middle-income countries, which incur more than USD 110 million annually in healthcare expenses and productivity losses. These circumstances highlight the necessity for stringent food safety and sanitation protocols in all food establishments, including bakeries, which serve as readily accessible sources of daily sustenance. P.D. 856 serves as the primary legal framework for this study due to its specific standards on food handling, personal hygiene, equipment sanitation, pest control, waste management, and sanitary inspections. Despite this established regulation, many small- and medium-scale bakeries continue to face challenges in achieving full compliance because of limited resources, aging facilities, and insufficient regulatory monitoring, which may lead to gaps in sanitation and inconsistent food safety practices.

International studies show that similar issues are observed in other countries. Research in Ghana and Poland found that while many bakeries follow basic hygiene routines, weaknesses in waste disposal, limited use of HACCP, and inconsistent enforcement still pose risks to consumers (Lawal et al., 2022; Drozd, 2022). Studies in Nigeria and Ethiopia likewise reported that inadequate facilities, lack of training, and irregular inspections contribute to unsafe food handling (Fakayode et al., 2022; Tamiru et al., 2022). Muhammad & Galadima (2023) confirmed that poor sanitation

and lack of clean water were the two leading causes of microbial contamination in bread. Similar concerns were identified in Italy, where organizational issues and non-compliance affected bakery operations (Currò et al., 2024). In contrast, bakeries implementing HACCP, ISO 22000, and structured cleaning programs showed improved food safety and product quality (Bukhovets et al., 2022; Sahani, 2024; Padmavathi & Praveen, 2023). Despite these developments, many studies emphasize that gaps remain between food handlers' knowledge and their actual practices, underscoring the need for continuous training and supervision (Nurlena et al., 2023).

Studies in the Philippines reflect similar challenges. Research involving bakeries, karinderyas, and food stalls reported inconsistencies in hygiene and sanitation due to limited space, outdated facilities, and a lack of regular monitoring (Cardona et al., 2024; Ann et al., 2023). Studies conducted in Batangas, Marawi, and Zamboanga del Norte documented microbial contamination and unsafe handling practices among small food operators and street vendors (Tolentino et al., 2024; Pontino, 2024; Solon, 2022). Similar issues were observed in restaurants, where inconsistent cleaning routines and irregular facility maintenance were noted (Sarmiento & Apritado, 2022; Escario, 2025). Consumer studies also show that customers value clean surroundings, proper handling, and visible compliance with sanitation rules (Rivas et al., 2023). Although food handlers in areas like Isabela, Ifugao, and Marikina City generally demonstrate good awareness of food safety principles, their actual practices remain moderate due to limited training or outdated equipment (Lucas et al., 2025; Shane et al., 2025).

Despite existing food safety regulations in the Philippines, sanitation compliance in small and medium food establishments remains a public health concern. Unsafe food can cause more than 200 diseases and may lead to long-term disability or death, with over 17,000 cases of acute bloody diarrhea reported in the Philippines in 2018 (WHO, 2019). Bakeries are particularly vulnerable due to frequent manual food handling and daily consumer exposure. However, limited

research has documented specific sanitation areas, such as personnel hygiene, equipment and utensil handling, facility cleanliness, and sanitation procedures, in selected local bakeries within selected municipalities of Agusan del Norte. This study is therefore necessary to assess bakery food safety and sanitation practices as a basis for strengthening public health protection and regulatory enforcement.

This research is significant to multiple stakeholders. Bakery owners and workers can use the findings to assess their sanitation practices and enhance compliance with national standards. Consumers may gain greater awareness of sanitation conditions, enabling them to make safer choices when purchasing baked products. Local government units and municipal health offices can use the results to improve monitoring systems and develop targeted training programs. Academic institutions, particularly the Department of Tourism and Hospitality Management, may benefit from the study as a reference for teaching, research, and community extension initiatives related to food safety and sanitation. Finally, the study may serve as a useful reference for future researchers, contributing to broader efforts to promote food safety and protect public health.

1.2 Theoretical Framework

This study is grounded on two established behavioral theories: The Health Belief Model (HBM) by Rosenstock (1977) and the Protection Motivation Theory (PMT) by Rogers (1975). These theories provide the theoretical basis for explaining bakery employees' food safety and sanitation practices and consumers' evaluation of these practices in selected Local Bakeries in Selected Municipalities of Agusan del Norte.

Health Belief Model (HBM) – Rosenstock (1977)

The Health Belief Model explains health-related behavior based on individuals' beliefs and perceptions regarding health risks and preventive actions. Its key constructs include perceived susceptibility, perceived severity, perceived benefits, and perceived barriers.

In this study, HBM is used to explain the food safety and sanitation practices of bakery employees, which serve as the independent variables. Employees' compliance with sanitation standards such as personal hygiene, proper handwashing, safe food handling, equipment cleaning, and waste management is influenced by their perception of susceptibility to food contamination, the severity of foodborne illnesses, the benefits of proper sanitation, and the barriers to compliance.

Bakery characteristics, including number of employees, years of operation, and business location, may influence the availability of sanitation facilities and workload, thereby affecting employees' perceived benefits and barriers. Employees who recognize both the health risks and operational consequences of poor sanitation are expected to demonstrate higher levels of food safety and sanitation compliance, which constitute part of the dependent outcome of bakery sanitation practices.

Protection Motivation Theory (PMT) – Rogers (1975)

The Protection Motivation Theory explains how individuals assess threats and decide on protective behaviors through threat appraisal and coping appraisal, including perceived severity, perceived vulnerability, response efficacy, self-efficacy, and response cost.

In this study, PMT is applied to explain consumer perception and evaluation of bakery food safety and sanitation, which also function as independent variables. Consumers assess potential health risks by observing visible sanitation indicators such as personnel hygiene, food handling behavior, cleanliness of the facility, and sanitation of utensils and equipment. Consumer characteristics such as age, sex, and educational attainment may influence how these sanitation practices are perceived and interpreted.

When consumers perceive that sanitation practices are effective in preventing food contamination, their confidence in food safety increases, contributing to positive evaluations of

bakery operations. Conversely, inadequate or poorly observed sanitation practices increase perceived risk and may negatively affect consumer trust and purchasing decisions. These consumer evaluations represent the dependent variable related to consumer perception of food safety and sanitation.

By integrating HBM and PMT, this study examines food safety and sanitation from both employee-based and consumer-based perspectives. The Health Belief Model explains how employees' beliefs influence their sanitation practices, while the Protection Motivation Theory explains how consumers evaluate and respond to these practices.

Food safety in bakeries is therefore influenced by the interaction of employee sanitation practices, consumer perception, and bakery characteristics. Together, these theories provide a comprehensive framework for understanding how behavioral and perceptual factors affect food safety and sanitation practices and consumer evaluations in local bakery operations.

Review of Related Literature

This section presents related literature and studies on food safety and sanitation practices in bakeries. It includes discussions on respondent profiles, bakery characteristics, and hygiene and sanitation procedures, as well as relevant research studies.

Food Safety Practices

Practices related to food safety, particularly in personal grooming and hygiene, play a critical role in ensuring the overall quality of food and protecting consumer health. Studies have shown that while food handlers are generally aware of hygiene standards, these standards are not always properly implemented. For instance, research in India and Vietnam revealed that food sellers often washed their hands and wore clean clothes, which are basic hygiene habits, but failed to use hairnets or wash their hands after handling money due to inadequate facilities and weak enforcement (Agrawal et al., 2025; Huynh-Van et al., 2022). In the Philippines, many eating houses were believed to follow sanitation regulations; however, shortages of water, protective gear, and supervision limited their ability to comply fully (Dollentas & Habla, 2021). Similarly, street and home-based sellers maintained some hygiene measures, yet laboratory tests showed bacterial contamination, highlighting a gap between perceived compliance and actual practice (Limon, 2021; Origenes et al., 2022). Studies conducted in Marawi and Zamboanga del Norte also reported that although food handlers possessed adequate knowledge, they struggled to meet sanitation standards due to insufficient training and infrastructure (Caye, 2024; Solon et al., 2022).

Food safety practices in food establishments largely depend on the personal hygiene and sanitation behaviors of food handlers. According to Bihag et al. (2021), personal hygiene practices such as regular handwashing before food handling, maintaining clean and appropriate work attire, proper grooming, use of personal protective equipment, and avoidance of behaviors that may contaminate food are essential in preventing the transfer of microorganisms during food

preparation. Food handlers who regularly monitor their health condition and consistently follow hygienic practices significantly reduce the risk of food contamination and foodborne illnesses, particularly in establishments where food is manually handled. In addition, sanitation practices play a crucial role in maintaining a safe food production environment. Bihag et al. (2021) further emphasized that routine cleaning and sanitizing of kitchen tools, equipment, utensils, and work areas, proper waste disposal, and safe food handling procedures help prevent environmental sources of contamination. Consistent sanitation practices, including minimizing direct hand contact with food and ensuring proper cooking at recommended temperatures, are necessary to maintain food safety standards in bakeries with daily consumer exposure. Together, personal hygiene and sanitation practices serve as fundamental components of effective food safety management and are widely recognized as key factors in ensuring safe and contamination-free food products.

Moreover, education, training, and strong management support have consistently been shown to enhance workers' hygiene practices and overall compliance with food safety standards. Studies conducted in different settings have confirmed that employees who receive proper guidance and ongoing training maintain higher levels of cleanliness and sanitation (Tyler et al., 2023; Siddiky et al., 2024; Rifat et al., 2022). Similarly, restaurant staff who participate in regular food safety programs demonstrate superior hygiene and sanitation performance (Labović et al., 2023; Ermeje et al., 2025). These findings indicate that although food handlers recognize the importance of hygiene, sustainable compliance with food safety practices requires sufficient resources, consistent supervision, and institutional commitment to continuous training.

Sanitation Practices

Food sanitation is one of the issues in food safety, ensuring it is contaminant-free and maintaining the health of consumers. The research by Aulia et al. (2024) showed that although the

majority of food handlers had satisfactory hygiene knowledge, many still engaged in poor sanitation practices, indicating that knowing what to do does not necessarily translate into practice. Very similar findings were reported by Pratama, (2023) during their assessment of the port canteens at Kendari; there, good sanitary conditions were observed mainly when workers were appropriately trained and the environment was clean and well-equipped, demonstrating that both knowledge and environment matter. Vi M. et al. (2025) also noted in the Philippines that students with a better understanding of hygiene and sanitation are healthier, highlighting the role of sanitation habits in well-being.

On the contrary, Zamudioa & Cabanisasa, (2024) assessed the facility for smoked fish production. They discovered that, even with a high level of hygiene awareness, some operational routines were not consistently applied. All these studies indicate that food sanitation relies on training, knowledge, and the right environment for practices to take place, keeping food prepared and served safely and cleanly.

Customer Assessment

Customer perceptions of food service quality are closely connected to the food safety practices observed in an establishment. Picardal (2019) explained that customers usually evaluate sanitation based on three major areas: food service and personnel hygiene, sanitation in facilities and safety procedures, and utensils and equipment handling. These aspects include visible behaviors such as staff cleanliness, proper handling of baked products, the condition of the surroundings, and the cleanliness of the equipment used in daily operations. These indicators serve as the basis of how customers form judgments regarding the overall safety and cleanliness of a bakery.

Research shows that customer perceptions are strongly influenced by these visible hygiene cues. Sarmiento and Apritado (2022) reported that personnel hygiene, such as clean

uniforms, proper grooming, and hygienic food handling, significantly improves customer evaluations. They also found that facility sanitation, including clean dining and preparation areas, orderly surroundings, and well-maintained restrooms, helps build customer confidence in the establishment. Proper cleaning, maintenance, and use of utensils and equipment further affect customer satisfaction, which demonstrates how these three components work together in shaping customer perceptions.

Studies from other contexts support these findings. Wei (2021) noted that staff hygiene, sanitation of facilities, and proper handling of utensils play important roles in influencing customer satisfaction and trust. Owusu et al. (2021) found that clean facilities and effective waste management encourage repeat patronage because customers associate cleanliness with safety. Nicolosi et al. (2023) observed that health-conscious consumers tend to prefer bakeries that maintain visible hygiene practices. Nakpathom (2023) also reported that sanitation conditions, food hygiene standards, and the overall ambiance of an establishment influence customer loyalty.

The literature collectively shows that customers form their assessment of food safety and sanitation based on what they can clearly see. These indicators include the cleanliness of the bakery's surroundings, the hygienic appearance of staff, the organization of display counters, the appearance of products, and the absence of unpleasant odors or waste. Customer evaluation therefore extends beyond product quality and includes the establishment's visible level of hygiene, sanitation, and safety compliance.

Food Service & Personnel's Hygiene

Hygiene among food service personnel is essential for ensuring food safety and preventing foodborne illnesses, yet gaps in knowledge and actual practice remain widespread. Although many food handlers are aware of hygiene standards, proper implementation is often inconsistent because of limited training, insufficient resources, or weak supervision (Ibrahim & Yusuf, 2023). In

Kano, Nigeria, only 18.7 percent of street vendors and 26.0 percent of canteen handlers demonstrated good hygiene knowledge. Common lapses such as inadequate handwashing and irregular use of protective gear show the need for regular training, certification, and stronger regulatory monitoring.

Even when food handlers possess good knowledge and positive attitudes toward hygiene, their practices may still be inconsistent. In Ilorin, 62.5 percent of fish vendors reported favorable attitudes toward food safety, especially in avoiding cross-contamination and maintaining personal hygiene. However, many did not consistently wear gloves or masks, which are necessary for preventing contamination (Ghali-Mohammed et al., 2024). Factors such as age and household responsibilities also influenced hygiene behaviors, suggesting that training programs and supervision need to consider demographic differences among workers.

Compliance with hygiene protocols is also shaped by workplace culture and operational limitations. In hospital food service facilities in Limpopo Province, South Africa, staff members understood hygiene requirements but were unable to follow them consistently because of cultural practices like wearing jewelry and operational challenges such as staff shortages, lack of proper uniforms, and insufficient inspection systems (Mphasha et al., 2024). These conditions hindered proper adherence to basic hygiene procedures.

Customer confidence in food establishments is closely linked to the visible hygiene practices of food handlers. Proper grooming, clean uniforms, regular handwashing, the use of protective gear, and hygienic handling of food strongly influence consumer perceptions of cleanliness and safety. The reviewed studies collectively show that improving personnel hygiene requires more than knowledge. It requires continuous training, supportive workplace conditions, adequate supplies, and consistent enforcement of hygiene standards to ensure safe food-handling behaviors.

Sanitation in Facilities & Safety Procedures

Sanitation in facilities and proper safety procedures are essential in foodservice operations because they help maintain food quality and prevent contamination. The study of Nurlena et al. (2023) assessed the sanitation and hygiene practices of 50 pastry production facilities and focused on equipment cleanliness, water quality, and compliance with sanitation regulations. Although 68 percent of the establishments met minimum sanitation requirements, several remained weak in areas such as waste disposal, equipment cleaning, and staff hygiene training. The authors emphasized that continuous sanitation monitoring and regular staff training are necessary to ensure the safety and cleanliness of food facilities.

Differences in sanitation performance across countries also highlight the importance of consistent standards. Research by Na et al. (2024) comparing shared kitchens in Korea, Japan, and the United States revealed variations in sanitation guidelines, inspection systems, and documentation practices. Facilities with insufficient policies and limited training showed poorer hygiene outcomes. The study suggested that adopting uniform regulations, improving inspection processes, and strengthening training can enhance safety in food preparation areas.

Access to sanitation facilities also influences compliance with hygiene procedures. An investigation of 250 street food vendors in Pretoria found that those with limited sanitation resources and lower food safety knowledge were less likely to follow hygiene guidelines. Ndhlovu and Tabit (2024) explained that knowledge and access to facilities must be addressed together because improving one factor without the other leads to minimal impact on overall food safety.

Consistent with these insights, this study gives importance to visible aspects of facility sanitation that customers can directly observe. Clean display areas, organized shelves, proper waste disposal, odor control, and the general neatness of the service environment all influence customers' perceptions of safety. These visible indicators shape their assessment of an

establishment's sanitation performance and reflect how effectively food facilities apply sanitation and safety procedures.

Utensils & Equipment Handling

Proper handling and sanitization of utensils and equipment are essential components of safe food preparation because they prevent cross-contamination and reduce the risk of foodborne illness. A study on street food vendors in Calapan City showed that food handlers demonstrated generally good hygiene practices, particularly in cleaning and sanitizing equipment. Tyler et al. (2023) found a strong positive correlation (r = 0.631) between hygiene procedures and food safety levels, demonstrating that consistent cleaning of utensils and equipment contributes directly to safer food handling.

The importance of proper cleaning practices is also reflected in findings from restaurant settings. Research from Iran involving 28 food establishments revealed high contamination levels on food-contact surfaces such as chopping boards and countertops. Fallahizadeh et al. (2025) identified ATP bioluminescence testing as a more effective method than simple visual inspection for detecting hidden contamination. The notable reduction in contamination after washing highlighted the need for thorough equipment sanitation and regular hygiene training for staff.

Local studies also reinforce the value of proper utensil and equipment management. An assessment of canteen food handlers in the Philippines showed moderate knowledge and practices in utensil sanitation, with the most significant gaps found in temperature control and cross-contamination prevention. Maye et al. (2024) noted that training and experience were important factors in improving compliance with food safety protocols. Consistent with these findings, this research gives attention to visible aspects of utensil and equipment handling that customers can directly observe, including the cleanliness of service utensils such as tongs, trays,

and scoops, as well as display cases, countertops, and packaging tools. These indicators influence how customers assess the sanitation level of a bakery and the safety of the products being offered.

Age

Age influences how individuals understand and apply food safety practices, as maturity, experience, and exposure to food preparation responsibilities tend to develop over time. Differences in learning processes across age groups may result in varying levels of food safety knowledge and compliance. Yi and Choi (2025), in a study conducted in areas surrounding Seoul, reported that individuals aged 40 to 60 demonstrated significantly higher knowledge of food safety practices than younger groups, particularly in food storage, hygiene, and contamination prevention. In contrast, younger individuals showed limited awareness of critical issues such as mold growth and appropriate freezing methods, which are essential for preventing foodborne diseases.

Similar findings were reported in Baghdad, where Abu-Almaaly (2025) observed that individuals aged 18 to 29 exhibited lower compliance with established food safety standards compared to middle-aged and older adults. The study suggested that increased life experience and accumulated knowledge contribute to more consistent and responsible food-handling behaviors among older individuals. As people age, they tend to develop a stronger sense of responsibility and risk awareness, which may influence their adherence to food safety guidelines.

Evidence from Bangladesh further supports the role of age in shaping food safety literacy. Islam et al. (2022) found that adults possessed a better understanding of nutrition, sanitation, and safe food practices, whereas teenagers displayed notable gaps in both knowledge and behavior. These generational differences were attributed to practical experience and repeated exposure to food safety practices among older individuals, while younger individuals often relied on informal or less reliable sources of information.

Additionally, demographic studies in Saudi Arabia have reinforced the importance of age in influencing food safety attitudes and behaviors. Almutlaq et al. (2025) reported that age, along with education, significantly affected individuals' food safety knowledge and practices. Although younger populations have greater access to digital information, older individuals were more consistent in applying proper food safety procedures, suggesting that access to information alone does not guarantee safe practices.

Overall, the reviewed studies consistently indicate that age is an important factor influencing food safety knowledge, attitudes, and practices across different settings. These findings highlight the need to examine age-related differences when assessing food safety and sanitation practices, particularly in food establishments where both employee behavior and consumer evaluation may vary across age groups.

Educational Attainment

Educational attainment plays a critical role in shaping individuals' food safety awareness, nutritional habits, and hygiene practices. Higher education generally provides greater exposure to information on nutrition, public health, and food safety regulations, which encourages more responsible food handling and consumption. While numerous studies demonstrate a positive relationship between education and safe food practices, the strength of this association can vary depending on socioeconomic and cultural contexts. Understanding this relationship is essential for designing effective interventions that promote food safety across different populations.

Research in various international contexts highlights the influence of education on food choices and hygiene practices. In London, communities with higher levels of education were found to consume larger quantities of fruits, vegetables, and fish, whereas less educated communities favored bread, potatoes, and high-fat foods (Fard et al., 2021). This indicates that education can shape dietary decisions both directly and indirectly through community norms. Similarly, studies

in Romania revealed that individuals with higher education not only demonstrated better food safety knowledge but also adopted safer kitchen practices and positive attitudes toward hygiene (Mihalache et al., 2021). These findings underscore the pivotal role of education in promoting both awareness and practical application of food safety measures.

Further evidence supports the positive impact of education on food safety in diverse cultural and socioeconomic settings. In India, individuals with higher educational backgrounds were more likely to read and understand food labels, particularly for edible oils and fats, highlighting the role of education in improving label literacy and informed food choices (Paliwal et al., 2025). In Bangladesh, food vendors with higher education exhibited greater knowledge of hygiene and food safety practices compared to their less educated counterparts, suggesting that even basic education can improve awareness of foodborne infections and preventive measures (Islam et al., 2024). Similar patterns were observed in Ethiopia, where street vendors with higher education and proper training were more likely to practice safe hygiene, especially when supported by continuous monitoring and access to sanitation facilities (Negassa et al., 2023). In Spain, young people with higher education demonstrated healthier diets, lower body mass indices, and fewer unhealthy habits, including smoking and excessive consumption of sugary drinks (Sandri et al., 2024). Collectively, these studies indicate that education not only increases knowledge but also influences behaviors that promote food safety and overall health.

Education also produces lasting improvements in professional and organizational food safety practices. In Malaysia, continuous food safety education for hospitality students increased awareness and encouraged consistent hygiene practices (Ahmad et al., 2024). Similarly, structured education for bakery and pastry students in Laguna, Philippines, improved both practical skills and employability, demonstrating that education contributes to competency and career readiness in the food industry (Santos & Aquino, 2023). Beyond vocational training, long-term educational

interventions can transform organizational culture. A case study of a military food service revealed that personnel gradually progressed from merely following rules to embracing food safety as a core value, supported by ongoing education and leadership (Zanin et al., 2021). In Indonesia, differences in educational levels among university food handlers significantly affected knowledge, attitudes, and hygiene practices, although continuous training remained necessary to reinforce proper behaviors (Firdani, 2022). These examples highlight that education, when combined with training and supervision, is crucial for consistent and effective food safety practices in both institutional and professional settings.

Taken together, these studies demonstrate a strong and consistent relationship between educational attainment and food safety awareness, attitudes, and practices. Individuals with higher education are more knowledgeable about hygiene measures, more likely to store food safely, and more consistent in their safe food handling. Educational interventions, whether through formal instruction, public campaigns, or vocational training, have proven effective in promoting food safety across diverse contexts. Strengthening education, particularly among food handlers and consumers in local communities, can improve public health outcomes, enhance compliance with food safety regulations, and contribute to the development of a safer, more responsible food system. Based on this evidence, the present study includes educational attainment as a demographic variable to examine its potential impact on bakery employees' sanitation practices and consumers' evaluation of food safety.

Sex

Sex influences food safety knowledge, attitudes, and practices in food handling. According to Gebre et al. (2023), differences in roles, access to training, and responsibility affect how men and women acquire and apply food safety knowledge. Their study among fruit and vegetable handlers in Ethiopia showed that educational interventions improved knowledge in both sexes,

with a stronger effect observed among women, supporting the inclusion of sex as a relevant variable in food safety and sanitation studies.

In low- and middle-income countries, food safety responsibilities often fall disproportionately on women, particularly in household and informal food production. Research highlights that women face challenges, including limited access to training, technology, safe infrastructure, and decision-making authority, which increases their vulnerability to foodborne risks (Milovanova et al., 2024). Similarly, women in Ethiopia's dairy value chain play an indispensable yet often overlooked role, and resource constraints further expose them to contamination hazards (Garsow et al., 2022). These conditions indicate the need to address structural inequalities to improve food safety outcomes.

Gender differences also influence compliance and performance in professional food handling environments. A study examining male food handlers in China found that although many followed good practices, gaps remained in handwashing, food storage, and container use. Factors such as income, work experience, and training were associated with better compliance, whereas older age was linked to lower adherence (Chen, 2025). This suggests that male food handlers may benefit from targeted, skills-based interventions to address specific weaknesses in food safety practices.

Women often take primary responsibility for food handling at home and in informal sectors, facing barriers such as limited education, restricted decision-making power, and insufficient policy support. Men, by contrast, may have greater access to formal training and resources in professional food service settings but do not always exhibit consistent hygiene practices. Recognizing these gender-based differences is essential for designing interventions that improve compliance, knowledge, and safe food handling across all groups. Understanding how sex influences food safety practices allows for the development of more effective and inclusive

strategies. Gender-sensitive training, improved infrastructure, and equitable access to resources can empower both men and women to adopt safer practices. Addressing these differences in policy and practice can strengthen the overall food safety system and contribute to better public health outcomes.

Business Location

The location of a food establishment affects sanitation practices because access to water, waste management, and infrastructure varies widely across communities. Moges et al. (2024) reported that the physical site of a business, including access to infrastructure such as clean water, waste disposal, and proper shelter, affects food handlers' ability to maintain hygienic operations.

In the Philippines, water, sanitation, and hygiene conditions vary widely across local communities, with some barangays experiencing limited access to safe water and adequate sanitation structures, contributing to inconsistent hygiene practices among establishments in underserved areas (Molina et al., 2021).

Environmental and infrastructure-related challenges are also observed in other countries. Bhandari and Bhusal (2021) found that street food vendors in Pokhara, Kaski, struggled to maintain sanitation because of poor access to clean water and proper waste management systems, indicating that water supply directly influences compliance with hygiene standards. Furthermore, Desye et al. (2023) noted that food safety knowledge, attitudes, and practices among street food vendors in low- and middle-income countries are largely shaped by contextual and environmental conditions, including access to sanitation facilities, education, and income levels. Vendors operating in poorly developed areas often face structural challenges that limit their ability to maintain hygienic standards.

Similarly, Ndhlovu and Tabit (2024) reported that street food vending sites in Marabastad, Pretoria, exhibited inadequate sanitation and low compliance with food safety requirements,

mainly due to the lack of proper facilities and infrastructure. Chumo et al. (2023) also emphasized that informal or marginalized settlements experience greater health and sanitation risks because of insufficient basic services, weak institutional support, and unsafe environments. These studies collectively highlight that business location plays a critical role in influencing food safety practices, as establishments situated in well-regulated, adequately serviced, and properly equipped areas are more capable of implementing and sustaining effective hygiene and sanitation measures.

Number of Employees

The number of employees in a bakery plays an important role in ensuring food safety, product quality, and overall operational efficiency. A properly structured workforce allows tasks to be distributed effectively, provides better supervision, and supports consistent hygiene and contamination control throughout the production process. Research has shown that microbial safety and contamination prevention in bakeries depend not only on equipment and methods but also on the management of the production environment, which is influenced by the size of the workforce (Vermelho et al., 2024). A sufficient number of employees makes it easier to monitor critical points such as fermentation, ingredient handling, and storage, thereby reducing the risk of microbial contamination.

Workforce size also affects how well spoilage and contamination risks are managed. Factors such as storage temperature and mold formation during cooling are closely linked to staff availability (Chou et al., 2024). When there are too few employees, it becomes difficult to maintain cleanliness, inspect equipment, and ensure proper storage conditions. These gaps increase the likelihood of microbial growth and product spoilage. On the other hand, an adequate number of employees allows responsibilities to be shared efficiently, enables prompt responses to potential hazards, and supports consistent implementation of hygiene measures.

Employee performance and bakery efficiency are further influenced by the number of workers. Studies in South Korea found that employees who were satisfied with their work and committed to the organization performed better, which positively affected overall bakery operations and food safety (Kang, 2021). Similar patterns have been observed in the Philippines. Labao et al. (2023) reported that food handlers with greater capacity to manage their workload and maintain systematic hygiene routines demonstrated better compliance with food safety practices. Their study showed that when enough personnel are present to divide tasks effectively, hygiene performance improves, helping prevent lapses in food handling and sanitation.

The evidence indicates that workforce size is a key factor in maintaining bakery hygiene and product safety. A sufficient number of employees helps enforce sanitation protocols, manage storage and cooling processes properly, and respond quickly to contamination risks. In contrast, understaffing can lead to lapses in hygiene, extended exposure of food to unsafe conditions, and higher chances of bacterial growth. To address these challenges, bakeries must maintain adequate staffing levels while providing continuous training and proper organization of their workforce. Ensuring that employees are both sufficient in number and well-trained strengthens operational efficiency and serves as a critical safeguard for safe and high-quality bakery products.

Years of Operation

Years of operation influence food safety performance because businesses with more experience tend to develop established systems, routines, and compliance with sanitation standards. In most cases, long-standing companies have well-organized and well-documented systems for handling food, maintaining hygiene, and complying with regulations. The study of Al Banna et al. (2021) emphasized that longer work experience among meat handlers in Bangladesh significantly contributed to improved food safety knowledge and practices. Handlers with more than ten years of experience demonstrated higher awareness of hygiene protocols and better

implementation of safe handling behaviors than less experienced workers. These findings highlighted that experience, along with education and training, serves as a strong predictor of food safety compliance.

The researchers also noted that older food establishments, which had undergone repeated inspections and regulatory updates over the years, had a higher likelihood of meeting sanitary standards. These businesses developed stronger skills in managing compliance, addressing infrastructure issues, and sustaining operational consistency, all of which contribute to improved hygiene performance. Similar observations were reported in the Philippines. Armas and Moralde (2024) found that ambulant food vendors with longer years of operation developed more systematic routines, demonstrated stronger organizational practices, and were better equipped to meet sanitation requirements compared to newer vendors. Their results show that operational longevity strengthens resilience and adherence to food-safety expectations. Chen et al. (2025) further supported this argument in their study of Chinese catering venues, where workers with longer experience showed higher compliance with safety procedures such as timely handwashing, food quality inspections, and contamination prevention. This suggests that establishments operating for many years tend to have more trained employees and more established safety routines than newly opened businesses.

A related study in Almaty, Kazakhstan, reported that employment duration was positively associated with food safety knowledge, attitudes, and practices (Kulpiisova et al., 2025). Food handlers with longer work experience displayed better safety performance, indicating that continuous exposure to food-handling environments enhances familiarity with hygiene procedures and encourages responsible behavior. In the Philippine context, a study of restaurants in Laguna and nearby areas also noted that establishments with more years of operation generally had more stable sanitation practices and stronger operational systems, reflecting accumulated

experience in meeting industry standards and regulatory expectations (Streamlining Operations and Best Practices in the Restaurant Industry, 2023). However, the Kazakhstan study also noted that many handlers had less than one year of experience, which correlated with lower knowledge levels despite generally positive attitudes.

Together, these studies suggest that the number of years an establishment or its workers have operated directly influences their food safety performance. Operational longevity not only reflects business stability but also represents accumulated learning, routine formation, and repeated exposure to regulatory requirements. These factors collectively strengthen compliance with food safety protocols over time.

The reviewed literature shows that food safety and sanitation in bakeries are influenced by a combination of individual, organizational, and environmental factors. Personal characteristics such as age, educational attainment, and sex affect food safety knowledge, attitudes, and behaviors, with older and more educated individuals generally demonstrating better practices. Organizational factors, including business location, workforce size, and years of operation, shape the ability of establishments to maintain hygienic conditions, provide adequate supervision, and develop consistent routines that support sanitation. Studies consistently report that proper food safety and sanitation practices depend not only on knowledge but also on training, resources, and supportive working conditions. Visible aspects of hygiene, such as personnel cleanliness, facility sanitation, and utensil and equipment handling, strongly influence customer assessment, which plays an important role in shaping trust and perceived safety. Overall, the literature highlights that food safety in bakeries results from the combined effects of employee behavior, organizational capacity, environmental conditions, and customer evaluation, all of which must work together to ensure safe and reliable food products.

1.4 Schematic Diagram

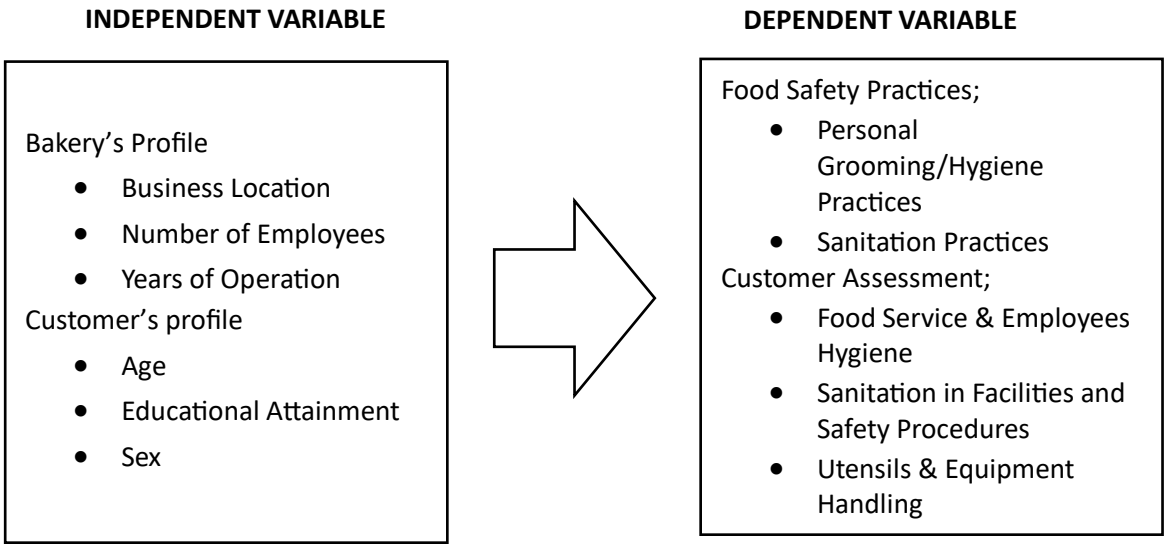


Figure 1 *shows the conceptual flow illustrating how the bakery and consumer profiles influence food safety and sanitation practices.*

1.5 Statement of the Problem

This study aims to assess the food safety and sanitation practices of selected local bakeries in the selected municipalities of Agusan del Norte. Specifically, it seeks to answer the following questions:

1. What is the profile of the selected local bakeries in terms of:

   1.1 business location:

   1.2 number of employees; and

   1.3 years of operation?

2. What is the profile of the customers of the selected local bakeries in terms of:

   2.1 age;

   2.2 educational attainment; and

   2.3 sex?

3. To what extent are food safety and sanitation practices implemented in the selected local bakeries, in terms of:

   3.1 personal grooming/hygiene practices; and

   3.2 sanitation practices?

4. What is the level of customers' assessment of food safety and sanitation practices in the selected local bakeries, in terms of:

   4.1 food service and personnel's hygiene;

   4.2 sanitation in facilities and safety procedures; and

   4.3 utensils and equipment handling?

5. Is there a significant relationship between the profile of respondents and the food safety and sanitatioYESn practices in the selected local bakeries?

1.6 Hypothesis

Null Hypotheses (H_o)

- H_o: There is no significant relationship between the profile of respondents (business and customer profiles) and the food safety and sanitation practices in selected local bakeries in the selected municipalities of Agusan del Norte.

1.7 Scope and Limitation

This study will focus on assessing the food safety and sanitation practices of registered bakeries in the selected municipalities of Agusan del Norte that have been operating for at least five (5) years to ensure stable and consistent operations. Participation will be voluntary, and only individuals who consent to take part will be included; those who decline will not be forced or counted. The study will exclude unregistered or newly established bakeries and will not include full demographic profiling of bakery staff, as the focus is on establishment-level factors influencing food safety and sanitation. Respondents under 18 years old will also be excluded to ensure that

only legally responsible consumers participate. For bakery respondents, only one representative per establishment, such as the owner, manager, supervisor, or the staff member responsible for food safety or food handling, will be included. Other employees will not participate. The findings will reflect practices and consumer perceptions during the data collection period and may not represent bakeries outside the selected areas.

1.8 Definition of Terms

For clarity and consistency, the following terms are defined as they are used, measured, and applied in this study.

Age. Customer's age group, as self-reported in the survey, categorized into 18–29, 30–39, 40–49, and 50+.

Bakery Respondent. The individual representing the bakery who completed the bakery survey, specifically the owner, manager, supervisor, or staff responsible for sanitation.

Business Location. Classification of the bakery as rural or urban, determined using the bakery profile checklist.

Customer Assessment. Customers' evaluation of bakery hygiene and sanitation, measured using the consumer survey instrument.

Educational Attainment. Highest level of education completed by the customer, as reported in the survey.

Food Safety. Compliance of bakeries with national hygiene and sanitation standards, measured through responses from both bakery and consumer instruments.

Food Safety Practices. Actions performed by bakery staff to maintain cleanliness and prevent contamination, measured via the bakery checklist.

Food Service and Personnel Hygiene. Customers' assessment of staff hygiene, including use of hairnets, aprons, gloves, bread covering, and possession of sanitary permits, measured through the consumer instrument.

Local Bakeries. Selected registered bakeries located in the three municipalities under study, anonymized as Municipality 1, Municipality 2, and Municipality 3..

Number of Employees. Total number of staff in each bakery at the time of data collection, as reported in the bakery survey.

Personal Grooming and Hygiene Practices. Bakery staff adherence to personal cleanliness, proper attire, and hand hygiene, measured via the bakery checklist.

Policy Enhancement. Recommendations for improving bakery sanitation monitoring and food safety procedures, developed from the study findings.

Sanitation in Facilities and Safety Procedures. Customers' evaluation of bakery facility cleanliness, adequacy of lighting and space, and garbage management, measured through the consumer instrument.

Sanitation Practices. Routine cleaning and sanitizing of surfaces, utensils, and equipment, as well as proper food handling by bakery staff, measured in the bakery checklist.

Sex. Customer's biological classification, male or female, as reported in the survey.

Utensils and Equipment Handling. Customers' evaluation of cleanliness, maintenance, and proper use of bakery utensils and equipment, measured through the consumer survey.

Years of Operation. Number of years the bakery has been operating at the time of data collection, as reported in the bakery survey.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter will discuss the research design and description of research respondents, instruments, ethical standards, data gathering procedure, and statistical treatment use.

2.1 Research Design

This study will employ a descriptive–quantitative research design using survey questionnaires to examine food safety and sanitation practices in local bakeries, as well as consumer perceptions of these practices. This design is appropriate because it assesses and measures the current situation using numerical data, providing an accurate picture of existing practices. Structured questionnaires allow the collection of reliable and measurable information, which can be analyzed statistically to determine compliance levels and consumer opinions. The results of this study can then serve as a basis for recommending improvements to food safety and sanitation practices in local bakeries.

2.2 Research Materials and Methods

Two survey questionnaires will be used in this study: one for bakery representatives and one for consumers.

The bakery questionnaire is a modified instrument based on the study of Bihag et al. (2021), *"Food Safety Practices Among Restaurants in Municipality of Buenavista, Agusan del Norte."* It does not include a profiling section for bakery staff because only one representative per establishment, such as the owner, manager, supervisor, or staff responsible for food safety, will complete the survey. Only bakery-level information is collected. The questionnaire consists of two parts. The first part gathers the bakery's demographic profile, including the number of employees, years of operation, and business location. The second part contains 20 statements on food safety

practices covering personal grooming, hygiene, and sanitation. Responses are measured using a four-point Likert scale: 4 – Strongly Agree, 3 – Agree, 2 – Disagree, and 1 – Strongly Disagree.

The consumer questionnaire is adapted from Picardal et al. (2019), *"Safety and Sanitation Procedures of Bakery Personnel in Selected Bakeries Along PUP Manila."* Minor adjustments were made to the demographic section to suit the target population, including age 18 years and above. This questionnaire also consists of two parts. The first part collects demographic information such as age, sex, and educational attainment. The second part includes 11 statements evaluating consumers' perceptions of food service and personnel hygiene, facility sanitation, safety procedures, and utensil and equipment handling. Responses are measured using a five-point Likert scale: 5 – Excellent, 4 – Above Average, 3 – Average, 2 – Below Average, and 1 – Very Poor.

The bakery questionnaire will be answered by the designated representative responsible for food safety to ensure that responses accurately reflect the operational practices of the establishment even if the owner is unavailable.

Both instruments will undergo expert validation to ensure clarity, relevance, and appropriateness. A pilot test will then be conducted to determine reliability using Cronbach's Alpha. Using separate questionnaires for bakery representatives and consumers is expected to provide accurate and comprehensive data on food safety and sanitation practices in selected local bakeries.

The main variables of the study include food safety and sanitation practices among bakeries and consumer perception of these practices. Indicators for the bakery questionnaire include personal grooming, employee hygiene, food handling, and sanitation procedures. Indicators for the consumer questionnaire include personnel hygiene, facility sanitation, safety procedures, and utensil and equipment handling.

2.3 Research Environment, Population, and Sample

This study will examine the food safety and sanitation practices of registered local bakeries listed in the official municipal records of the selected study areas in Agusan del Norte, coded as Municipality 1, Municipality 2, and Municipality 3. The focus will be on key areas such as personal hygiene, food handling, cleaning and sanitation, storage practices, and waste management to assess how these establishments maintain proper standards in their daily operations.

The respondents will include bakery owners, managers, or designated bakery representatives, as well as consumers aged 18 years and above. Prior to data collection, participants will be oriented on the study's purpose, objectives, and potential benefits and risks. They will be informed of their right to refuse or withdraw from participation at any time without consequences, and written informed consent will be obtained to ensure voluntary participation.

Survey questionnaires for bakery respondents will be administered on-site at their establishments using purposive sampling. Consumer respondents will be selected through convenience sampling at the selected bakery locations during weekends to capture typical customer flow. Survey sessions will be conducted from 6:00–9:00 AM for morning customers and 3:00–5:00 PM for afternoon customers, with each respondent expected to spend approximately 30 minutes completing the survey.

Table 1 presents the distribution of bakery and consumer respondents included in the study.

RESPONDENT	**Population**	**Sample**	
	F	**f**	**%**
MUNICIPALITY 1			
BAKESHOP 1	50	10	3.16
BAKESHOP 2	12	2	0.63
BAKESHOP 3	150	29	9.18
BAKESHOP 4	20	4	1.27
BAKESHOP 5	20	4	1.27
BAKESHOP 6	50	10	3.16

BAKESHOP 7	40	8	2.53
BAKESHOP 8	50	10	3.16
BAKESHOP 9	10	2	0.63
BAKESHOP 10	30	6	1.90
BAKESHOP 11	50	10	3.16
BAKESHOP 12	290	56	17.72
BAKESHOP 13	25	5	1.58
MUNICIPALITY 2			
BAKESHOP 1	80	15	4.75
BAKESHOP 2	40	8	2.53
BAKESHOP 3	30	6	1.90
BAKESHOP 4	50	10	3.16
BAKESHOP 5	100	19	6.01
BAKESHOP 6	10	2	0.63
BAKESHOP 7	70	13	4.11
BAKESHOP 8	25	5	1.58
BAKESHOP 9	50	10	3.16
BAKESHOP 10	120	23	7.28
BAKESHOP 11	30	6	1.90
MUNICIPALITY 3			
BAKESHOP 1	10	2	0.63
BAKESHOP 2	60	12	3.80
BAKESHOP 3	100	19	6.01
BAKESHOP 4	50	10	3.16
TOTAL 28	**1622**	**316**	**100.00**

Table 1 *Distribution of Bakery and Consumer Respondents*

2.4 Research Data and Analysis Plan

The researchers will use purposive sampling for bakery respondents and convenience sampling for consumer respondents. This method is appropriate because it allows the inclusion of bakeries that meet specific criteria (i.e., registered and operating for at least five years) while ensuring accessibility to willing consumers.

Statistical Treatment

In presenting, analyzing, and interpreting the data, the following statistical tools will be used:

- **Frequency Count and Percentage** – to answer SOP 1 and SOP 2, determining the profile of bakery and consumer respondents.

- **Weighted Mean and Standard Deviation** – to answer SOP 3 and SOP 4 assessing the level of food safety and sanitation practices among local bakeries and the consumers' evaluations.
- **Pearson's Correlation Coefficient** – to determine the relationship between the respondents' profiles and their assessment of food safety and sanitation practices. This test is appropriate because the study examines the relationship between quantitative variables, such as age, years of operation, and hygiene or sanitation ratings, allowing the researchers to identify whether significant correlations exist among these variables.

2.5 Research Ethical Standards

A. Intellectual Property Rights

The study will respect all intellectual property rights by ensuring originality and proper acknowledgment of sources. All secondary data, references, and adapted research instruments will be appropriately cited to honor the intellectual property of other researchers. Respondent anonymity will also be maintained throughout the data collection and analysis process to protect both participants and the researchers' contributions.

B. Informed Consent

The respondents will be bakery staff, managers, or consumers from selected local bakeries in the study municipalities coded as Municipality 1, Municipality 2, and Municipality 3. Before data collection, the researchers will conduct a brief orientation to explain the study's purpose, objectives, and potential benefits and risks. Participants will be informed of their right to refuse or withdraw from participation at any time without any negative consequences. Written informed consent forms will be provided in a language understandable to the respondents (Filipino or English).

C. Data Privacy and Confidentiality

All personal data collected, including demographic information and survey responses, will be treated with strict confidentiality. Respondent identities will be anonymized and coded to protect privacy. Only the research team will have access to the dataset and identification codes. Data will be securely stored and used solely for academic purposes.

D. Voluntary Participation

Participation in the study will be entirely voluntary. Respondents will be informed that they can withdraw at any time without penalty. If a participant withdraws, all their data will be excluded from the study and securely deleted. No animals will be involved in this study; therefore, considerations for alternatives to animal use are not applicable.

E. Minimization of Harm

The researchers will ensure a safe, respectful, and professional environment during data collection. The study poses minimal risk to participants. Should any discomfort arise, respondents will be referred to appropriate support services such as the Municipal Health Office or the Guidance Office of Saint Michael College of Caraga. A debriefing will be provided after participation to ensure no emotional, psychological, or physical harm occurs.

F. Beneficence

The study will benefit the community by identifying gaps and challenges in food safety and sanitation practices in the selected municipalities. Findings may guide policymakers, business owners, and consumers in improving food handling practices and ensuring consumer safety. Respondents may request access to the study results after completion.

G. Justice and Fair Participant Selection

Respondents will be selected using fair and well-defined inclusion criteria, ensuring alignment with the study's objectives. Participants from diverse backgrounds will be included equitably, without discrimination based on sex, age, educational attainment, or social status.

H. Data Integrity and Accuracy

Data collection and participant selection will follow strict criteria. Only officially registered bakeries that have operated for at least five years and consumers who regularly purchase bakery products will be included. Data will be recorded, encoded, and cross-checked to ensure accuracy and reliability, minimizing errors in analysis.

I. Transparency in Reporting

Study findings will be reported honestly, without manipulation or misrepresentation of data. All sources will be appropriately cited and acknowledged to respect intellectual property rights and uphold ethical research standards. Any use of AI tools in summarizing statements or drafting the report will be verified for accuracy by the research team, with no conflicts of interest involved.

J. Use of Patented or Copyrighted Materials

No patented materials will be involved in this study. All copyrighted materials, including academic references and adapted instruments, will be properly cited in accordance with intellectual property laws. The researchers will ensure that all sources are acknowledged to maintain academic integrity.

K. Considerations for Human Subjects

All human participants will be treated in accordance with national and institutional ethical standards. Respondents will be informed about potential risks and benefits, and written informed consent will be obtained. If any emotional, physical, or psychological concerns arise during data collection, appropriate support services will be made available.

L. Responsible Use of AI and Related Technology

Digital tools and AI, such as ChatGPT, Grammarly, and research databases (Google Scholar, Elicit, Typeset.io), will be responsibly used to support research accuracy, summarization, and clarity. All outputs will be verified by the researchers to ensure credibility and integrity of the study.

M. Ethical Approval

Ethical approval will be obtained from the Institutional Review Board of Saint Michael College of Caraga to confirm compliance with all relevant guidelines regarding respondent protection, data privacy, and intellectual property rights.

2.6 Data Gathering Procedure

To gather data, the researchers will first secure endorsement letters addressed to the Municipal Mayor's Office and the Municipal Business Permits and Licensing Office (BPLO) of the study municipalities, coded as Municipality 1, Municipality 2, and Municipality 3, to obtain official lists of registered bakeries. In addition, formal letters will be sent to each bakery to seek their approval and permission to participate in the study. Once the lists and approvals are obtained, the survey questionnaires for bakery owners, managers, staff, and consumers will be finalized. The instruments will undergo expert validation to ensure clarity, relevance, and accuracy, followed by a pilot test to determine reliability using Cronbach's Alpha.

Upon approval, the survey will be administered to all listed bakeries using purposive sampling for bakery respondents and convenience sampling for consumers aged 18 years and above. Consumers will be approached at selected bakery locations during weekends to capture typical customer flow. Survey sessions will be conducted from 6:00–9:00 AM for morning customers and 3:00–5:00 PM for afternoon customers, with each respondent expected to spend approximately 30 minutes completing the survey. Bakery respondents will complete the survey on-site, while consumers will participate voluntarily during the designated periods.

After data collection, all responses will be tallied, encoded, and submitted to the statistician for analysis to ensure that the findings accurately reflect bakery food safety and sanitation practices as well as consumer assessments in the selected municipalities.

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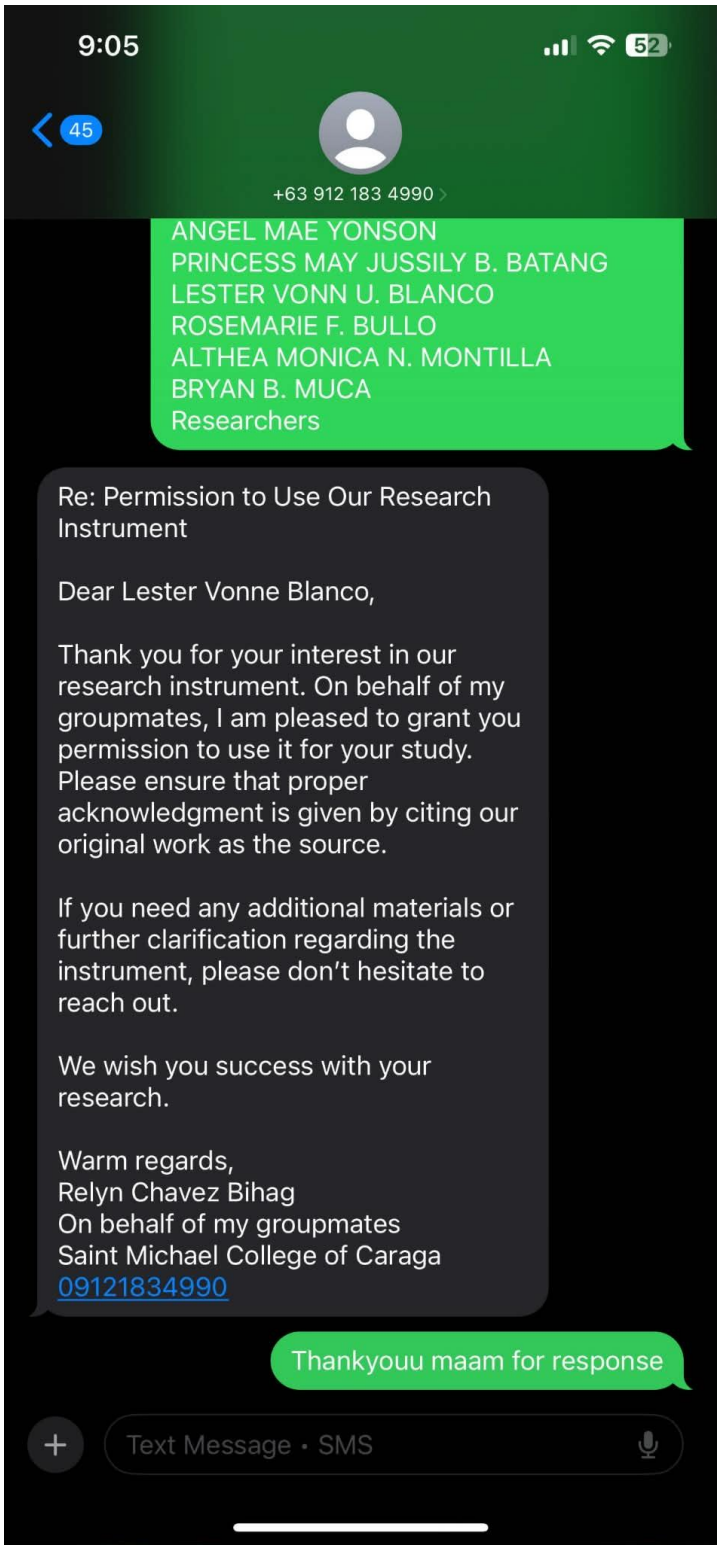
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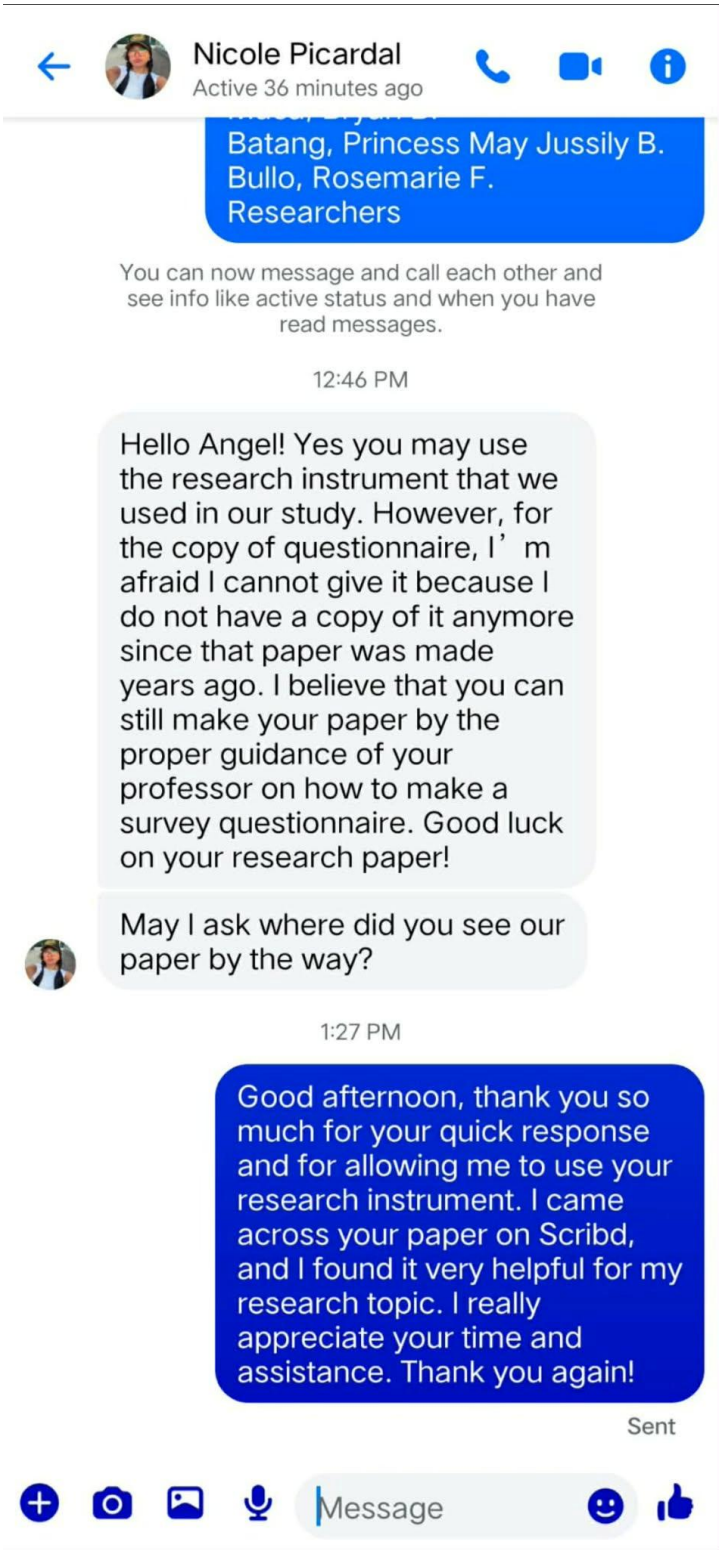
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APPENDICES


9:05
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+63 912 183 4990
ANGEL MAE YONSON
PRINCESS MAY JUSSILY B. BATANG
LESTER VONN U. BLANCO
ROSEMARIE F. BULLO
ALTHEA MONICA N. MONTILLA
BRYAN B. MUCA
Researchers
Re: Permission to Use Our Research Instrument
Dear Lester Vonne Blanco,
Thank you for your interest in our research instrument. On behalf of my groupmates, I am pleased to grant you permission to use it for your study. Please ensure that proper acknowledgment is given by citing our original work as the source.
If you need any additional materials or further clarification regarding the instrument, please don't hesitate to reach out.
We wish you success with your research.
Warm regards,
Relyn Chavez Bihag
On behalf of my groupmates
Saint Michael College of Caraga
09121834990
Thankyouu maam for response
Text Message • SMS


Nicole Picardal
Active 36 minutes ago
Batang, Princess May Jussily B.
Bullo, Rosemarie F.
Researchers
You can now message and call each other and see info like active status and when you have read messages.
12:46 PM
Hello Angel! Yes you may use the research instrument that we used in our study. However, for the copy of questionnaire, I'm afraid I cannot give it because I do not have a copy of it anymore since that paper was made years ago. I believe that you can still make your paper by the proper guidance of your professor on how to make a survey questionnaire. Good luck on your research paper!
May I ask where did you see our paper by the way?
1:27 PM
Good afternoon, thank you so much for your quick response and for allowing me to use your research instrument. I came across your paper on Scribd, and I found it very helpful for my research topic. I really appreciate your time and assistance. Thank you again!
Sent
Message

CURRICULUM VITAE



NAME: ANGEL MAE YONSON

ADDRESS: D-2 SANTA ANA, NASIPIT, AGUSAN DEL NORTE

EMAIL: Angelmae_yonson@smccnasipit.edu.ph

Personal Information

Sex: FEMALE

Date of birth: AUGUST 11, 2005

Height: 4'11

Weight: 39 kg

Civil Status: SINGLE

Educational Attainment

Elementary: SANTA ANA ELEMENTARY SCHOOL

Secondary: SAINT MICHAEL COLLEGE OF CARAGA

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
LOCAL SUPREME GOVERNMENT	VICE PRESIDENT
SLICE	LADY OF THE HOUSE
SLICE	HEAD OF COMMUNICATION

CURRICULUM VITAE



NAME: PRINCESS MAY JUSSILY B. BATANG

ADDRESS: D-11 MACOPA CARMEN, AGUSAN DEL NORTE

EMAIL: Princessmayjussily_batang@smccnasipit.edu.ph

Personal Information

Sex: FEMALE

Date of birth: MAY 6, 2004

Height: 5'0

Weight: 50kg

Civil Status: SINGLE

Educational Attainment

Elementary: CARMEN CENTRAL ELEMENTARY SCHOOL

Secondary: OUR LADY OF CARMEN ACADAMY OF CARAGA INCORPORATED

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
SLICE	MEMBER

CURRICULUM VITAE



NAME: LESTER VONNE U. BLANCO

ADDRESS: Purok 2 BARANGAY 3, BUENAVISTA, AGUSAN DEL NORTE

EMAIL: Lestervonne_blanco@smccnasipit.edu.ph

Personal Information

Sex: MALE

Date of birth: JANUARY 9, 2005

Height: 5’9

Weight: 73 kg

Civil Status: SINGLE

Educational Attainment

Elementary: MALAYBALAY CITY CENTRAL SCHOOL

Secondary: BUKIDNON NATIONAL HIGH SCHOOL

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
LOCAL SUPREME GOVERNMENT	BUSINESS MANAGER
SLICE	MEMBER

CURRICULUM VITAE



NAME: ROSEMARIE F. BULLO

ADDRESS: P-2 CAPON CARMEN, AGUSAN DEL NORTE

EMAIL: rosemarie_bullo@smccnasipit.edu.ph

Personal Information

Sex: FEMALE

Date of birth: FEBRARY 18, 2003

Height: 4’9

Weight: 45kg

Civil Status: SINGLE

Educational Attainment

Elementary: VALENCIA ELEMENTARY SCHOOL

Secondary: SENIOR HIGH SCHOOL CARMEN

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	**POSITIONS HELD**
SLICE	MEMBER

CURRICULUM VITAE



NAME: ATHEA MONICA N. MONTILLA

ADDRESS: 122 CURVERA STREET NASIPIT, AGUSAN DEL NORTE

EMAIL: altheamonicamontill641@gmail.com

Personal Information

Sex: FEMALE

Date of birth: JULY 12, 2005

Height: 4'11

Weight: 52kg

Civil Status: SINGLE

Educational Attainment

Elementary: NASIPIT CENTRAL ELEMENTARY SCHOOL

Secondary: SAINT MICHAEL COLLEGE OF CARAGA

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	**POSITIONS HELD**
SLICE	MEMBER

CURRICULUM VITAE



NAME: BRYAN B. MUCA

ADDRESS: D-11 TAPNIGUE SANGAY BUENAVISTA, AGUSAN DEL NORTE

EMAIL: bryan_muca@smccnasipit.edu.ph

Personal Information

Sex: MALE

Date of birth: SEPTEMBER 25, 2004

Height: 5’8

Weight: 80kg

Civil Status: SINGLE

Educational Attainment

Elementary: TAPNIGUE ELEMENTARY SCHOOL

Secondary: SANGAY NATIONAL HIGH SCHOOL

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	**POSITIONS HELD**
SLICE	MEMBER